LARGE VALUE ESTIMATES FOR DIRICHLET POLYNOMIALS AND TRILINEAR SUMS OVER SHORT INTERVALS

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Abstract. Recently, Matomäki, Pandey, Pliego, Teräväinen and Wang [8] improved the Baker–Harman–Pintz [2] estimate of trilinear sums over short intervals of length x^θ, which naturally reduces to estimating mean values of products of three Dirichlet polynomials. Based on recent works of Guth and Maynard [4] and Chen, Gupta and Li [3] on new large value estimates for Dirichlet polynomials, they improved the Baker–Harman–Pintz estimate for $\theta > \frac{3}{5}$. In this paper, we use their method to improve the Baker–Harman–Pintz estimate for $\frac{1}{2} < \theta < \frac{4}{7}$ and get new results for all $\frac{1}{2} < \theta < 1$. Our result yields improvements on many problems, including the work of Li [7] on primes in short intervals of length $x^{0.52}$.

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Notation

Throughout this paper, we always assume that x is a sufficiently large integer, ε is a sufficiently small positive constant and $K \geqslant 1$. The appearance of K in the exponent of a logarithm will always signify that the result holds for every $K \geqslant 1$ with an implied constant that depends on K. We shall ignore the presence of the integer multiples of ε in many places below. The letter p, with or without subscript, is reserved for prime numbers. Let c and c_i ($i \in \mathbb{N}$) denote positive constants which may have different values at different places. We use both Landau and Vinogradov notations, and implied constants may depend on fixed parameters. We write $m \sim M$ to mean that $M \leqslant m < 2M$ and $m \asymp M$ to mean that $c_1 M < m < c_2 M$. Let $\frac{1}{2} < \theta < 1$, $y = x^{\theta+\varepsilon}$, $y_1 = x\exp\big(-3(\log x)^{1/3}\big)$, $T = x^{1-\theta}$ and $T_0 = \exp\big((\log x)^{1/3}\big)$. We use $M(s)$, $N(s)$, $R(s)$ and some other capital letters, with or without subscript, to denote some divisor-bounded Dirichlet polynomials

$$M(s) = \sum_{m\sim M} a_m m^{-s}, \quad N(s) = \sum_{n\sim N} b_n n^{-s}, \quad R(s) = \sum_{r\sim R} c_r r^{-s}.$$

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Note that for a divisor-bounded Dirichlet polynomial $M(s)$, there exists c_d such that for every $d \in \mathbb{N}$,

$$\sum_{m \leqslant M} |a_m|^{2d} \ll M(\log M)^{c_d}.$$

We say a Dirichlet polynomial $M(s)$ is *prime-factored* if we have

$$\left| M\left(\frac{1}{2}+it\right)\right| \ll M^{\frac{1}{2}}(\log x)^{-K}$$

for $T_0 < |t| < T$. In fact, this holds when a_m is the characteristic function of primes or of numbers with a bounded number of prime factors restricted to certain ranges. For example, if we have

$$a_m = \sum_{m=p_1\cdots p_k} 1$$

and the least prime factor of m is $\gg \exp\left((\log x)^{4/5}\right)$, then we know that $M(s)$ is prime-factored by [[5], Lemma 1.5].

1. Introduction

In this paper we seek the mean value estimates of products of three Dirichlet polynomials in the following form:

$$\int_{T_0}^{T} \left| M\left(\frac{1}{2}+it\right) N\left(\frac{1}{2}+it\right) R\left(\frac{1}{2}+it\right)\right| \ll x^{\frac{1}{2}}(\log x)^{-K}, \tag{1}$$

where $M, N, R \gg x^{\varepsilon}$ and $MNR \asymp x$. This type of mean value estimates plays a crucial role in finding primes in short intervals. In fact, by [[2], Lemma 6] or [[5], Lemma 7.2], (1) implies the following asymptotic formula for trilinear sums:

$$\sum_{\substack{mnr\in[x-y,x)\\ m\sim M\\ n\sim N}} a_m b_n c_r = \frac{y}{y_1} \sum_{\substack{mnr\in[x-y_1,x)\\ m\sim M\\ n\sim N}} a_m b_n c_r + O\left(y(\log x)^{-K}\right). \tag{2}$$

By various asymptotic formulas of the form (2), together with a combinatorial decomposition of prime counting function (usually done by applying Heath-Brown identity or Harman's sieve), one would get an asymptotic formula or a nontrivial lower bound for the number of primes in a short interval $[x-y, x)$.

In 1996, Baker, Harman and Pintz [1] proved (1) under the condition that $R(s)$ is prime-factored and M, N lie in specific regions. Write $M = x^{\alpha_1}$ and $N = x^{\alpha_2}$, then their restrictions on M, N becomes

$$|\alpha_1 - \alpha_2| < 2\theta - 1, \qquad 1 - \alpha_1 - \alpha_2 < \gamma(\theta), \tag{3}$$

where

$$\gamma(\theta) = \min\left(4\theta - 2,\ \frac{20\theta - 9}{11},\ \frac{72\theta - 37}{11}\right).$$

Later in 2001, they [2] further generalize this result by replacing $\gamma(\theta)$ with a more flexible expression:

$$\gamma(\theta) = \max_{g\in\mathbb{N}} \gamma_g(\theta), \quad \gamma_g(\theta) = \min\left(4\theta - 2,\ \frac{(8g-4)\theta - (4g-3)}{4g-1},\ \frac{24g\theta - (12g+1)}{4g-1}\right).$$

Taking $g = 3$, this reduces to their 1996 result. Using this 2001 result (with $g = 5$) as the main Type-II arithmetic information input to give asymptotic formulas of the form (2), together with an application of Harman's sieve, they showed that there is always a prime number in the short interval $[x - y, x]$, where $\theta = 0.525$. In 2025, Li [7] improved this to $\theta = 0.52$, and he also used this 2001 result (with $g = 6$) to produce asymptotic formulas of the form (2).

The proof of the 2001 Baker–Harman–Pintz result uses large value estimates for Dirichlet polynomials, but only classical bounds from classical mean value estimate and Halász–Montgomery–Huxley large value estimate were available at that time. In 2026, Guth and Maynard [4] and Chen, Gupta and Li [3] proved new large value estimates for Dirichlet polynomials, which is better than classical bounds in some cases. Recently, Matomäki, Pandey, Pliego, Teräväinen and Wang [8] improved the 2001 Baker–Harman–Pintz result for $\theta > \frac{3}{5}$ by developing a new method and inserting those new large value estimates into the proof. In this paper, by applying the method developed in [8], we obtain an improvement over the 2001 Baker–Harman–Pintz result for $\frac{1}{2} < \theta < \frac{4}{7}$.

Theorem 1. *Let $\frac{1}{2} < \theta < \frac{4}{7}$. Write $M = x^{\alpha_1}$ and $N = x^{\alpha_2}$. Then (1) holds if $R(s)$ is prime-factored and*

$$|\alpha_1 - \alpha_2| < 2\theta - 1, \qquad 1 - \alpha_1 - \alpha_2 < \gamma(\theta),$$

where

$$\gamma(\theta) = \max\left(\max_{g_1\in\mathbb{N}} \gamma^1_{g_1}(\theta), \max_{\substack{g_2\in\mathbb{N}\\ g_2\geqslant 5\\ 2\theta-1\in I_{g_2}}} \gamma^2_{g_2}(\theta),\ \gamma^3_{g_3}(\theta)\right),$$

$$\gamma^1_{g_1}(\theta) = \min\left(4\theta - 2,\ \frac{(8g_1-4)\theta-(4g_1-3)}{4g_1-1},\ \frac{24g_1\theta-(12g_1+1)}{4g_1-1}\right),$$

$$\gamma^2_{g_2}(\theta) = \frac{(4g_2-1)(2\theta-1)}{2g_2-1} - \varepsilon, \quad I_{g_2} = \left[\max\left(\frac{1}{4g_2+6}, \frac{5}{24g_2+5}\right), \frac{2g_2-1}{8g_2^2+4g_2-2}\right],$$

$$\gamma^3_{g_3}(\theta) = \min\left(\frac{(4g_3-3)\theta-(2g_3-2)}{2g_3-1}, \frac{5((8g_3-1)\theta-4g_3)}{4g_3-5}, \frac{(4g_3-1)(2\theta-1)}{2g_3-1}\right) - \varepsilon,$$

$$g_3 = g_3(\theta) = \max\left(2,\ \left\lfloor\frac{5(2-3\theta)}{12(2\theta-1)}\right\rfloor + 1\right).$$

Remark. The term $\gamma^1(\theta)$ is the same as $\gamma(\theta)$ in the 2001 Baker–Harman–Pintz result. When $\theta = 0.52$, the 2001 Baker–Harman–Pintz result $\gamma_{\mathrm{BHP}}(\theta) = \gamma^1_6(\theta)$ gives $\frac{2}{25} = 0.08$, while our new $\gamma(\theta)$ gives $\frac{19}{225} - \varepsilon \approx 0.0844$, giving a 5.5% improvement.

Apart from results with conditions similar to (3), we also get the following result with a different type of conditions.

Theorem 2. *Let $\frac{1}{2} < \theta < 1$. Write $M = x^{\alpha_1}$ and $N = x^{\alpha_2}$. Then (1) holds if $R(s)$ is prime-factored and any of the following 3 conditions holds:*

$$(i) \qquad |\alpha_1 - \alpha_2| < 2\theta - 1, \qquad \max(\alpha_1, \alpha_2) \geqslant 1 - \theta, \qquad \alpha_1 + \alpha_2 \geqslant \frac{1+3\theta}{4} + \varepsilon;$$

$$(ii) \qquad |\alpha_1 - \alpha_2| < 2\theta - 1, \qquad \frac{5(1-\theta)}{6} \leqslant \min(\alpha_1, \alpha_2) \leqslant \max(\alpha_1, \alpha_2) \leqslant 1 - \theta,$$

and, for some $g \in \mathbb{N}$ *such that* $2 \leqslant g \leqslant \max\left(2, \lceil \frac{1-\theta}{2\theta-1} \rceil\right)$,

$$\alpha_1+\alpha_2 > \max\left(\delta_g^{1,1},\ \min\left(\delta_g^{1,2},\ \max\left(\delta_g^{1,3},\ \delta_g^{1,4}\right)\right)\right)+\varepsilon \quad \textit{or} \quad \delta_g^{1,1}+\varepsilon < \alpha_1+\alpha_2 < \delta_g^{1,5}-\varepsilon,$$

where

$$\begin{aligned}
\delta_g^{1,1} &= \delta_g^{1,1}(\theta) = \frac{(4g-3)(1-\theta)-|\alpha_1-\alpha_2|}{2g-1},\\
\delta_g^{1,2} &= \delta_g^{1,2}(\theta) = \frac{8g-1-(12g-1)\theta-|\alpha_1-\alpha_2|}{2g-1},\\
\delta_g^{1,3} &= \delta_g^{1,3}(\theta) = \frac{24g-5-(40g-5)\theta-5|\alpha_1-\alpha_2|}{4g-5},\\
\delta_g^{1,4} &= \delta_g^{1,4}(\theta) = \frac{6g-2-(8g-2)\theta-|\alpha_1-\alpha_2|}{2g-1},\\
\delta_g^{1,5} &= \delta_g^{1,5}(\theta) = (4g-1)\theta-2g+1+|\alpha_1-\alpha_2|;
\end{aligned}$$

(iii) $\quad |\alpha_1-\alpha_2| < 2\theta-1, \quad \dfrac{5(1-\theta)}{8} \leqslant \min(\alpha_1,\alpha_2) \leqslant \dfrac{5(1-\theta)}{6}, \quad \max(\alpha_1,\alpha_2) \leqslant 1-\theta,$

and, for some $g \in \mathbb{N}$ *such that* $2 \leqslant g \leqslant \max\left(2, \lceil \frac{1-\theta}{2\theta-1} \rceil\right)$,

$$\alpha_1+\alpha_2 > \max\left(\delta_g^{2,1},\ \min\left(\delta_g^{2,2},\ \max\left(\delta_g^{2,3},\ \delta_g^{2,4}\right)\right)\right)+\varepsilon \quad \textit{or} \quad \delta_g^{2,1}+\varepsilon < \alpha_1+\alpha_2 < \delta_g^{2,5}-\varepsilon,$$

where

$$\begin{aligned}
\delta_g^{2,1} &= \delta_g^{2,1}(\theta) = \frac{(10g-5)(1-\theta)-|\alpha_1-\alpha_2|}{5g-1},\\
\delta_g^{2,2} &= \delta_g^{2,2}(\theta) = \frac{10g(2-3\theta)-|\alpha_1-\alpha_2|}{5g-1},\\
\delta_g^{2,3} &= \delta_g^{2,3}(\theta) = \frac{4g(3-5\theta)-|\alpha_1-\alpha_2|}{2g-1},\\
\delta_g^{2,4} &= \delta_g^{2,4}(\theta) = \frac{30g-5-(40g-5)\theta-2|\alpha_1-\alpha_2|}{2(5g-1)},\\
\delta_g^{2,5} &= \delta_g^{2,5}(\theta) = 5g(2\theta-1)+|\alpha_1-\alpha_2|.
\end{aligned}$$

There are many interesting applications of our Theorem 1 and Theorem 2. For example, by replacing the function $\gamma_{\mathrm{BHP}}(\theta)$ used in [7] with our new $\gamma(\theta)$ in Theorem 1, we can improve the exponent $\theta = 0.52$ concerning primes in short intervals obtained in [7]. The upper and lower bounds for the functions $\mathbf{LB}(\theta)$ and $\mathbf{UB}(\theta)$ defined in [7] can also be improved.

Theorem 3. *There exists a computable number* $c > 0$ *such that, for all sufficiently large* x, *the interval* $[x - x^{0.52-c}, x]$ *contains prime numbers.*

2. The Matomäki–Pandey–Pliego–Teräväinen–Wang Method

In this section we briefly introduce the new method of obtaining (1) developed by Matomäki, Pandey, Pliego, Teräväinen and Wang [8] and state some results we need in the proof of Theorem 1 and Theorem 2. Let $\mathcal{T} \subseteq [T, 2T]$. Define $\mathcal{N}_M(\nu; \mathcal{T})$ as the maximal cardinality of a

1-separated subset $\mathcal{U} \subseteq \mathcal{T}$ such that

$$\left|M\left(\frac{1}{2}+it\right)\right| \geqslant M^{\frac{1}{2}-\nu} \quad \text{for all } t \in \mathcal{U}.$$

By applying the classical mean value estimate [[6], Theorem 9.7], Halász–Montgomery–Huxley large value estimate [[6], Corollary 9.9], Guth–Maynard large value estimate [[4], Theorem 1.1] and Chen–Gupta–Li large value estimate [[3], Theorem 1.1] to $M^k(s)$, Matomäki, Pandey, Pliego, Teräväinen and Wang [[8], Lemma 2.3(i)] obtained the following upper bound for $\mathcal{N}_M(\nu;\mathcal{T})$ that holds for every fixed $k \in \mathbb{N}$:

$$\mathcal{N}_M(\nu;\mathcal{T}) \ll x^{\mathcal{K}(k\alpha,\nu;\theta)+o(1)}, \tag{4}$$

where

$$\mathcal{K}(\alpha,\nu;\theta) = \min\left(1-\theta,\ \max\left(2\alpha\nu,\ \mathcal{K}_0(\alpha,\nu;\theta)\right)\right), \tag{5}$$

$$\begin{aligned}\mathcal{K}_0(\alpha,\nu;\theta) = \ &\min\left(1-\theta+\alpha(6\nu-2),\ 1-\theta+\alpha(2\nu-1),\right.\\ &\left.\max\left(1-\theta+\alpha\left(4\nu-\frac{8}{5}\right),\ \frac{1-\theta}{2}+\alpha(4\nu-1)\right)\right).\end{aligned} \tag{6}$$

Lemma 2.1. *Let $M_j = x^{\alpha_j}$. Suppose that $\alpha_1+\alpha_2+\alpha_3 = 1$ and $\alpha_1,\alpha_2,\alpha_3 \geqslant \varepsilon$. Suppose also that one of $M_1(s)$, $M_2(s)$ and $M_3(s)$ is prime-factored. For $j = 1,2,3$, define the function $P_j : [0,1-\theta] \to [0,\infty]$ as*

$$P_j(t) = \inf\left\{\alpha_j\nu : \nu \in [\varepsilon^6,\varepsilon^{-1}], \min_{1\leqslant k\leqslant \lceil\varepsilon^{-7}\rceil} \mathcal{K}(k\alpha_j,\nu;\theta) \geqslant t\right\}.$$

If we have

$$P_1(t)+P_2(t)+P_3(t) > t+\varepsilon^{10}$$

for every $t \in [0,1-\theta]$, then we have

$$\int_{\mathcal{T}} \left|M_1\left(\frac{1}{2}+it\right)M_2\left(\frac{1}{2}+it\right)M_3\left(\frac{1}{2}+it\right)\right| \ll x^{\frac{1}{2}}(\log x)^{-K}.$$

Proof. This is [[8], Proposition 3.2] with $J = 3$, $\tau = 1-\theta$ and

$$H_j(\nu) = \min_{1\leqslant k\leqslant \lceil\varepsilon^{-7}\rceil} \mathcal{K}(k\alpha_j,\nu;\theta).$$

□

By a standard dyadic partition together with Lemma 2.1, we can reduce the problem of proving (1) to the task of showing a piecewise-linear function $P_1(t)+P_2(t)+P_3(t)-t-\varepsilon^{10}$ is positive. In order to give a lower bound for $P_j(t)$, we define the following auxiliary function $\mathcal{R}_\alpha$ as

$$\mathcal{R}_\alpha(t) = \alpha\inf\left\{\nu : \nu \geqslant 0,\ \mathcal{K}(\alpha,\nu;\theta) \geqslant t\right\}. \tag{7}$$

Since $\mathcal{K}(\alpha,\nu;\theta)$ has an explicit expression, we can also write $\mathcal{R}_\alpha(t)$ explicitly as

$$\begin{aligned}\mathcal{R}_\alpha(t) = \ &\max\left(0,\ \min\left(\frac{t}{2},\ \max\left(\frac{t-(1-\theta)+2\alpha}{6},\ \frac{t-(1-\theta)+\alpha}{2},\right.\right.\right.\\ &\left.\left.\left.\min\left(\frac{t-(1-\theta)+\frac{8}{5}\alpha}{4},\ \frac{t-\frac{1-\theta}{2}+\alpha}{4}\right)\right)\right)\right).\end{aligned} \tag{8}$$

Moreover, we have

$$\mathcal{R}_\alpha(t)=\begin{cases}0, & \varepsilon\leqslant\alpha\leqslant\frac{1-\theta}{2} \text{ and } 0\leqslant t\leqslant 1-\theta-2\alpha,\\ \frac{t-(1-\theta)+2\alpha}{6}, & \varepsilon\leqslant\alpha\leqslant\frac{1-\theta}{2} \text{ and } 1-\theta-2\alpha\leqslant t\leqslant 1-\theta-\frac{4}{5}\alpha,\\ \frac{t-(1-\theta)+\frac{8}{5}\alpha}{4}, & \varepsilon\leqslant\alpha\leqslant\frac{1-\theta}{2} \text{ and } 1-\theta-\frac{4}{5}\alpha\leqslant t\leqslant 1-\theta-\frac{2}{5}\alpha,\\ \frac{t-(1-\theta)+\alpha}{2}, & \varepsilon\leqslant\alpha\leqslant\frac{1-\theta}{2} \text{ and } 1-\theta-\frac{2}{5}\alpha\leqslant t\leqslant 1-\theta,\\ \frac{t}{2}, & \frac{1-\theta}{2}\leqslant\alpha\leqslant\frac{5(1-\theta)}{6} \text{ and } 0\leqslant t\leqslant\alpha-\frac{1-\theta}{2},\\ \frac{t-(1-\theta)+2\alpha}{6}, & \frac{1-\theta}{2}\leqslant\alpha\leqslant\frac{5(1-\theta)}{6} \text{ and } \alpha-\frac{1-\theta}{2}\leqslant t\leqslant 1-\theta-\frac{4}{5}\alpha,\\ \frac{t-(1-\theta)+\frac{8}{5}\alpha}{4}, & \frac{1-\theta}{2}\leqslant\alpha\leqslant\frac{5(1-\theta)}{6} \text{ and } 1-\theta-\frac{4}{5}\alpha\leqslant t\leqslant 1-\theta-\frac{2}{5}\alpha,\\ \frac{t-(1-\theta)+\alpha}{2}, & \frac{1-\theta}{2}\leqslant\alpha\leqslant\frac{5(1-\theta)}{6} \text{ and } 1-\theta-\frac{2}{5}\alpha\leqslant t\leqslant 1-\theta,\\ \frac{t}{2}, & \frac{5(1-\theta)}{6}\leqslant\alpha\leqslant 1-\theta \text{ and } 0\leqslant t\leqslant\alpha-\frac{1-\theta}{2},\\ \frac{t-\frac{1-\theta}{2}+\alpha}{4}, & \frac{5(1-\theta)}{6}\leqslant\alpha\leqslant 1-\theta \text{ and } \alpha-\frac{1-\theta}{2}\leqslant t\leqslant\frac{3(1-\theta)}{2}-\alpha,\\ \frac{t-(1-\theta)+\alpha}{2}, & \frac{5(1-\theta)}{6}\leqslant\alpha\leqslant 1-\theta \text{ and } \frac{3(1-\theta)}{2}-\alpha\leqslant t\leqslant 1-\theta,\\ \frac{t}{2}, & \alpha>1-\theta.\end{cases}\tag{9}$$

Now, by the definitions of $P_j(t)$ and $\mathcal{R}_\alpha(t)$, we can easily deduce that

$$P_j(t)=\max_{1\leqslant k\leqslant\lceil\varepsilon^{-7}\rceil}\left(\frac{1}{k}\mathcal{R}_{k\alpha_j}(t)\right).\tag{10}$$

Thus, we can use $\mathcal{R}_\alpha(t)$ to give lower bounds for $P_j(t)$. We also note that for $t\leqslant\frac{\varepsilon^6}{2}$, we have $P_1(t)+P_2(t)+P_3(t)\geqslant(\alpha_1+\alpha_2+\alpha_3)\varepsilon^6=\varepsilon^6$ and $P_1(t)+P_2(t)+P_3(t)-t-\varepsilon^{10}\geqslant\frac{\varepsilon^6}{2}-\varepsilon^{10}>0$. Hence we can assume that $t>\frac{\varepsilon^6}{2}$.

Looking at the slopes of $\mathcal{R}_\alpha(t)$ as in [8], we can prove some elementary lemmas that will be helpful in the proof of Theorem 1 and Theorem 2.

Lemma 2.2. *Suppose that $\alpha_1+\alpha_2+\alpha_3=1$ and $\alpha_1,\alpha_2,\alpha_3\geqslant\varepsilon$. For any $\frac{1}{2}<\theta<1$, suppose that there exists an integer $g\geqslant 1$ such that*

$$\frac{5(1-\theta)}{6}\leqslant g\alpha_3\leqslant\alpha_1\leqslant\alpha_2\leqslant 1-\theta.$$

Assume that

$$(2g-1)\alpha_3<(4g-1)(2\theta-1)+|\alpha_1-\alpha_2|-\varepsilon^7.$$

Then we have

$$P_1(t)+P_2(t)+P_3(t)-t>\varepsilon^{10}$$

for every $t\in[\frac{\varepsilon^6}{2},\ 1-\theta]$.

Proof. By (10), we have

$$P_1(t)+P_2(t)+P_3(t)-t\geqslant\mathcal{R}_{\alpha_1}(t)+\mathcal{R}_{\alpha_2}(t)+\frac{1}{g}\mathcal{R}_{g\alpha_3}(t)-t=F_1(t).$$

Since $g\alpha_3\leqslant\alpha_1\leqslant\alpha_2$, we have

$$g\alpha_3-\frac{1-\theta}{2}\leqslant\alpha_1-\frac{1-\theta}{2}\leqslant\alpha_2-\frac{1-\theta}{2}\leqslant\frac{3(1-\theta)}{2}-\alpha_2\leqslant\frac{3(1-\theta)}{2}-\alpha_1\leqslant\frac{3(1-\theta)}{2}-g\alpha_3.$$

The slopes of the piecewise-linear function $F_1(t)$ in the 7 intervals, given by $\frac{\varepsilon^6}{2}$, $1-\theta$ and the above 6 intermediate values, are

$$\frac{1}{2g}, \quad \frac{1}{4g}, \quad -\frac{1}{4}+\frac{1}{4g}, \quad -\frac{1}{2}+\frac{1}{4g}, \quad -\frac{1}{4}+\frac{1}{4g}, \quad \frac{1}{4g}, \quad \frac{1}{2g}$$

respectively. Hence $F_1(t)$ is non-decreasing in $[\frac{\varepsilon^6}{2},\ \alpha_1-\frac{1-\theta}{2}]$, non-increasing in $[\alpha_1-\frac{1-\theta}{2},\ \frac{3(1-\theta)}{2}-\alpha_1]$ and non-decreasing again in $[\frac{3(1-\theta)}{2}-\alpha_1,\ 1-\theta]$. Since

$$\begin{aligned} F_1\left(\frac{\varepsilon^6}{2}\right) &= \mathcal{R}_{\alpha_1}\left(\frac{\varepsilon^6}{2}\right)+\mathcal{R}_{\alpha_2}\left(\frac{\varepsilon^6}{2}\right)+\frac{1}{g}\mathcal{R}_{g\alpha_3}\left(\frac{\varepsilon^6}{2}\right)-\frac{\varepsilon^6}{2} \\ &= \left(\frac{1}{2}+\frac{1}{2}+\frac{1}{2g}-1\right)\frac{\varepsilon^6}{2}=\frac{\varepsilon^6}{4g}>\varepsilon^{10}, \end{aligned}$$

we only need to check the value of $F_1\left(\frac{3(1-\theta)}{2}-\alpha_1\right)$. Write $t_0=\frac{3(1-\theta)}{2}-\alpha_1$. Now,

$$\begin{aligned} F_1(t_0) &= \mathcal{R}_{\alpha_1}(t_0)+\mathcal{R}_{\alpha_2}(t_0)+\frac{1}{g}\mathcal{R}_{g\alpha_3}(t_0)-t_0 \\ &= \frac{(4g-1)(2\theta-1)-(2g-1)\alpha_3+(\alpha_2-\alpha_1)}{8g} \\ &= \frac{(4g-1)(2\theta-1)-(2g-1)\alpha_3+|\alpha_1-\alpha_2|}{8g}>\frac{\varepsilon^7}{8g}>\varepsilon^{10}, \end{aligned}$$

since $\alpha_2 \geqslant \alpha_1$. □

Lemma 2.3. *Suppose that $\alpha_1+\alpha_2+\alpha_3=1$ and $\alpha_1,\alpha_2,\alpha_3 \geqslant \varepsilon$. For any $\frac{1}{2}<\theta<1$, define*

$$w(\alpha)=\begin{cases} 1-\theta-\frac{2}{5}\alpha, & \frac{5(1-\theta)}{8} \leqslant \alpha \leqslant \frac{5(1-\theta)}{6}, \\ \frac{3(1-\theta)}{2}-\alpha, & \frac{5(1-\theta)}{6} \leqslant \alpha \leqslant 1-\theta. \end{cases}$$

Suppose that

$$\frac{5(1-\theta)}{8} \leqslant \alpha_1 \leqslant \alpha_2 \leqslant 1-\theta.$$

Assume that there exists an integer $k \geqslant 2$ such that

$$\frac{1}{k}\mathcal{R}_{k\alpha_3}(w(\alpha_1))>\frac{\alpha_3-(2\theta-1)}{2}+\varepsilon^7.$$

Then we have

$$P_1(t)+P_2(t)+P_3(t)-t>\varepsilon^{10}$$

for every $t \in [\frac{\varepsilon^6}{2},\ 1-\theta]$. Moreover, the same result holds if

$$\alpha_1 \geqslant \frac{5(1-\theta)}{8}, \quad \alpha_2 \geqslant 1-\theta, \quad |\alpha_1-\alpha_2|<2\theta-1 \quad \textit{and} \quad \alpha_3 \leqslant \frac{3(1-\theta)}{4}-\varepsilon^5.$$

Proof. Assume first that $\frac{5(1-\theta)}{8} \leqslant \alpha_1 \leqslant \alpha_2 \leqslant 1-\theta$. By (10), we have

$$P_1(t)+P_2(t)+P_3(t)-t \geqslant \mathcal{R}_{\alpha_1}(t)+\mathcal{R}_{\alpha_2}(t)+\frac{1}{k}\mathcal{R}_{k\alpha_3}(t)-t=F_2(t).$$

For $t \in [\frac{\varepsilon^6}{2},\ \alpha_1 - \frac{1-\theta}{2}]$, we have $\mathcal{R}_{\alpha_1}(t) = \mathcal{R}_{\alpha_2}(t) = \frac{t}{2}$, and by (10) we know that

$$P_3(t) \geqslant \frac{1}{\lceil \frac{1-\theta}{\varepsilon} \rceil} \mathcal{R}_{\lceil \frac{1-\theta}{\varepsilon} \rceil \alpha_3}(t) = \frac{t}{2\lceil \frac{1-\theta}{\varepsilon} \rceil} > \varepsilon^8 > \varepsilon^{10}$$

since $\alpha_3 \geqslant \varepsilon$. Hence $P_1(t) + P_2(t) + P_3(t) - t \geqslant \varepsilon^{10}$ for $t \in [\frac{\varepsilon^6}{2},\ \alpha_1 - \frac{1-\theta}{2}]$. For $t \in [\alpha_1 - \frac{1-\theta}{2},\ w(\alpha_1)]$, the slope of $\mathcal{R}_{\alpha_1}(t)$ is $\leqslant \frac{1}{4}$, the slope of $\mathcal{R}_{\alpha_2}(t)$ is $\leqslant \frac{1}{2}$, and the slope of $\frac{1}{k}\mathcal{R}_{k\alpha_3}(t)$ is $\leqslant \frac{1}{2k} \leqslant \frac{1}{4}$, hence $F_2(t)$ is non-increasing in $[\alpha_1 - \frac{1-\theta}{2},\ w(\alpha_1)]$. For $t \in [w(\alpha_1),\ 1-\theta]$, both $\mathcal{R}_{\alpha_1}(t)$ and $\mathcal{R}_{\alpha_2}(t)$ have slope $\frac{1}{2}$, hence $F_2(t)$ is non-decreasing in $[w(\alpha_1),\ 1-\theta]$. Thus, we only need to check the value of $F_2(w(\alpha_1))$. Now,

$$\begin{aligned} F_2(w(\alpha_1)) &= \mathcal{R}_{\alpha_1}(w(\alpha_1)) + \mathcal{R}_{\alpha_2}(w(\alpha_1)) + \frac{1}{k}\mathcal{R}_{k\alpha_3}(w(\alpha_1)) - w(\alpha_1) \\ &= w(\alpha_1) - (1-\theta) + \frac{\alpha_1 + \alpha_2}{2} + \frac{1}{k}\mathcal{R}_{k\alpha_3}(w(\alpha_1)) - w(\alpha_1) \\ &= \frac{(2\theta - 1) - \alpha_3}{2} + \frac{1}{k}\mathcal{R}_{k\alpha_3}(w(\alpha_1)) > \varepsilon^7 > \varepsilon^{10} \end{aligned}$$

by (9) and our assumption. Now the proof of the first part of Lemma 2.3 is completed.

Assume next that $\alpha_2 \geqslant 1 - \theta$. If $\alpha_1 \geqslant 1 - \theta$, then we have $\mathcal{R}_{\alpha_1}(t) = \mathcal{R}_{\alpha_2}(t) = \frac{t}{2}$, and by (10) we know that

$$P_3(t) \geqslant \frac{1}{\lceil \frac{1-\theta}{\varepsilon} \rceil} \mathcal{R}_{\lceil \frac{1-\theta}{\varepsilon} \rceil \alpha_3}(t) = \frac{t}{2\lceil \frac{1-\theta}{\varepsilon} \rceil} > \varepsilon^8 > \varepsilon^{10}$$

since $\alpha_3 \geqslant \varepsilon$. Hence $P_1(t) + P_2(t) + P_3(t) - t \geqslant \varepsilon^{10}$ in this case. Now we can assume that $\alpha_1 < 1 - \theta$. Since $|\alpha_1 - \alpha_2| < 2\theta - 1$ and $\alpha_3 \leqslant \frac{3(1-\theta)}{4}$, we have

$$\alpha_1 = \frac{1 - \alpha_3 - |\alpha_1 - \alpha_2|}{2} > \frac{1 - \alpha_3 - 2\theta + 1}{2} = 1 - \theta - \frac{\alpha_3}{2} \geqslant \frac{5(1-\theta)}{8}.$$

By the above discussions, we only need to check the value of $F_2(w(\alpha_1))$. Now,

$$\begin{aligned} F_2(w(\alpha_1)) &= \mathcal{R}_{\alpha_1}(w(\alpha_1)) + \mathcal{R}_{\alpha_2}(w(\alpha_1)) + \frac{1}{k}\mathcal{R}_{k\alpha_3}(w(\alpha_1)) - w(\alpha_1) \\ &= \frac{w(\alpha_1) - (1-\theta) + \alpha_1}{2} + \frac{w(\alpha_1)}{2} + \frac{1}{k}\mathcal{R}_{k\alpha_3}(w(\alpha_1)) - w(\alpha_1) \\ &= \frac{\alpha_1 - (1-\theta)}{2} + \frac{1}{k}\mathcal{R}_{k\alpha_3}(w(\alpha_1)) = G(\alpha_1). \end{aligned}$$

Note that $w(\alpha)$ has a slope $-\frac{2}{5}$ or -1, for fixed k, θ, α_3 and any $1 - \theta - \frac{\alpha_3}{2} \leqslant z_1 < z_2 \leqslant 1 - \theta$ we have

$$G(z_2) - G(z_1) \geqslant \frac{z_2 - z_1}{2} - \frac{w(z_1) - w(z_2)}{2k} \geqslant \frac{k-1}{2k}(z_2 - z_1) > 0.$$

Now we only need to find an integer $k \geqslant 2$ such that $G\left(1 - \theta - \frac{\alpha_3}{2}\right) > \varepsilon^7$. We consider the following 3 cases:

(1). $\varepsilon \leqslant \alpha_3 < \frac{1-\theta}{3}$. Now $1 - \theta - \frac{\alpha_3}{2} > \frac{5(1-\theta)}{6}$, and

$$w\left(1 - \theta - \frac{\alpha_3}{2}\right) = \frac{3(1-\theta)}{2} - \left(1 - \theta - \frac{\alpha_3}{2}\right) = \frac{1 - \theta + \alpha_3}{2}.$$

Write $l = \lfloor \frac{1-\theta}{\alpha_3} \rfloor$. Taking $k = l \geqslant 3$. Then

$$\begin{aligned} G\left(1-\theta-\frac{\alpha_3}{2}\right) &\geqslant -\frac{\alpha_3}{4} + \frac{w\left(1-\theta-\frac{\alpha_3}{2}\right)-(1-\theta)+l\alpha_3}{2l} \\ &= -\frac{\alpha_3}{4} + \frac{1-\theta+\alpha_3-2(1-\theta-l\alpha_3)}{4l} \\ &\geqslant \frac{-l\alpha_3+1-\theta+\alpha_3-2(1-\theta-l\alpha_3)}{4l} \\ &= \frac{\alpha_3-(1-\theta-l\alpha_3)}{4l}. \end{aligned}$$

Taking $k = l+1 \geqslant 4$. Since $l+1 \geqslant \lceil \frac{1-\theta}{\alpha_3} \rceil$, we have $\mathcal{R}_{k\alpha_3}\left(w\left(1-\theta-\frac{\alpha_3}{2}\right)\right) = \frac{w\left(1-\theta-\frac{\alpha_3}{2}\right)}{2}$. Then

$$\begin{aligned} G\left(1-\theta-\frac{\alpha_3}{2}\right) &= -\frac{\alpha_3}{4} + \frac{1-\theta+\alpha_3}{4(l+1)} \\ &= \frac{1-\theta-l\alpha_3}{4(l+1)}. \end{aligned}$$

Combining the two lower bounds, we have

$$\begin{aligned} G\left(1-\theta-\frac{\alpha_3}{2}\right) &\geqslant \frac{1}{2l+1}\left(l\frac{\alpha_3-(1-\theta-l\alpha_3)}{4l} + (l+1)\frac{1-\theta-l\alpha_3}{4(l+1)}\right) \\ &= \frac{\alpha_3}{4(2l+1)}. \end{aligned}$$

Since $l \leqslant \frac{1-\theta}{\alpha_3}$, we have

$$G\left(1-\theta-\frac{\alpha_3}{2}\right) \geqslant \frac{\alpha_3}{4(2l+1)} \geqslant \frac{\alpha_3^2}{4(2(1-\theta)+\alpha_3)} > \frac{\alpha_3^2}{12(1-\theta)} > \varepsilon^3 > \varepsilon^7.$$

(2). $\frac{1-\theta}{3} \leqslant \alpha_3 < \frac{1-\theta}{2}$. Now $\frac{3(1-\theta)}{4} < 1-\theta-\frac{\alpha_3}{2} \leqslant \frac{5(1-\theta)}{6}$, and

$$w\left(1-\theta-\frac{\alpha_3}{2}\right) = 1-\theta-\frac{2}{5}\left(1-\theta-\frac{\alpha_3}{2}\right) = \frac{3(1-\theta)+\alpha_3}{5}.$$

Since

$$G\left(1-\theta-\frac{\alpha_3}{2}\right) \geqslant -\frac{\alpha_3}{4} + \frac{1}{2}\mathcal{R}_{2\alpha_3}\left(w\left(1-\theta-\frac{\alpha_3}{2}\right)\right),$$

we only need to prove that

$$\mathcal{R}_{2\alpha_3}\left(w\left(1-\theta-\frac{\alpha_3}{2}\right)\right) > \frac{\alpha_3}{2} + \varepsilon^6.$$

By (9), this follows from the 3 conditions:

$$\begin{aligned} \frac{w\left(1-\theta-\frac{\alpha_3}{2}\right)}{2} - \frac{\alpha_3}{2} &= \frac{3(1-\theta)-4\alpha_3}{10} > \varepsilon^6, \\ \frac{w\left(1-\theta-\frac{\alpha_3}{2}\right)-(1-\theta)+\frac{16}{5}\alpha_3}{4} - \frac{\alpha_3}{2} &= \frac{7\alpha_3-2(1-\theta)}{20} > \varepsilon^6, \\ \frac{w\left(1-\theta-\frac{\alpha_3}{2}\right)-\frac{1-\theta}{2}+2\alpha_3}{4} - \frac{\alpha_3}{2} &= \frac{(1-\theta)+2\alpha_3}{40} > \varepsilon^6, \end{aligned}$$

since $\frac{2(1-\theta)}{7}+\varepsilon^5<\frac{1-\theta}{3}\leqslant\alpha_3<\frac{1-\theta}{2}<\frac{3(1-\theta)}{4}-\varepsilon^5$.

(3). $\frac{1-\theta}{2}\leqslant\alpha_3\leqslant\frac{3(1-\theta)}{4}-\varepsilon^5$. Now $\frac{5(1-\theta)}{8}<1-\theta-\frac{\alpha_3}{2}\leqslant\frac{3(1-\theta)}{4}$, and

$$w\left(1-\theta-\frac{\alpha_3}{2}\right)=1-\theta-\frac{2}{5}\left(1-\theta-\frac{\alpha_3}{2}\right)=\frac{3(1-\theta)+\alpha_3}{5}.$$

Now we have

$$\mathcal{R}_{2\alpha_3}\left(w\left(1-\theta-\frac{\alpha_3}{2}\right)\right)-\frac{\alpha_3}{2}=\frac{w\left(1-\theta-\frac{\alpha_3}{2}\right)}{2}-\frac{\alpha_3}{2}=\frac{3(1-\theta)-4\alpha_3}{10}>\varepsilon^6.$$

Combining the 3 cases, the proof of the second part of Lemma 2.3 is completed. □

3. Proof of Theorem 1 and Theorem 2

The γ^1 part of Theorem 1 follows directly from [[2], Lemma 9]. We first prove the γ^2 part of Theorem 1. Since we have

$$\frac{2g_2-1}{8g_2^2+4g_2-2}<\frac{1}{4g_2+4}$$

for all $g_2\in\mathbb{N}^+$, from $2\theta-1\in I_{g_2}$ we know that

$$\frac{1}{4g_2+6}\leqslant 2\theta-1\leqslant\frac{1}{4g_2+4}. \tag{11}$$

Since we have

$$\begin{aligned}2\theta-1\leqslant\frac{1}{4g_2+4}\implies 4\theta-2&\leqslant\frac{(4g_2+2)(2\theta-1)+1}{4g_2+3}\\&=\frac{(8(g_2+1)-4)\theta-(4(g_2+1)-3)}{4(g_2+1)-1}\end{aligned} \tag{12}$$

and

$$\begin{aligned}2\theta-1\geqslant\frac{1}{4g_2+6}\implies 4\theta-2&\leqslant\frac{12(g_2+1)(2\theta-1)-1}{4g_2+3}\\&=\frac{(24(g_2+1)\theta-(12(g_2+1)+1)}{4(g_2+1)-1},\end{aligned} \tag{13}$$

choosing $g=g_2+1$ in [[2], Lemma 9] gives Theorem 1 with all $\alpha_3\leqslant 4\theta-2$. Thus, we can assume that $4\theta-2<\alpha_3<\frac{(4g_2-1)(2\theta-1)}{2g_2-1}$. Without loss of generality, we suppose that $\alpha_1\leqslant\alpha_2$, while the case $\alpha_2\leqslant\alpha_1$ follows immediately by reversing the roles of α_1 and α_2. Since $|\alpha_1-\alpha_2|<2\theta-1$ and $\alpha_3>4\theta-2$, we have

$$\alpha_2=\frac{1-\alpha_3-\alpha_1+\alpha_2}{2}=\frac{1-\alpha_3+|\alpha_1-\alpha_2|}{2}<\frac{1-(2\theta-1)}{2}=1-\theta. \tag{14}$$

Since $2\theta-1\geqslant\frac{5}{24g_2+5}$ and $\alpha_3>4\theta-2$, we have

$$g_2\alpha_3>g_2(4\theta-2)\geqslant\frac{5(1-\theta)}{6}. \tag{15}$$

Since $2\theta-1\leqslant\frac{2g_2-1}{8g_2^2+4g_2-2}$, we have

$$\frac{(2g_2+1)(4g_2-1)(2\theta-1)}{2g_2-1}+(2\theta-1)\leqslant 1. \tag{16}$$

By $\alpha_3 < \frac{(4g_2-1)(2\theta-1)}{2g_2-1}$ and $|\alpha_1-\alpha_2| < 2\theta-1$, (16) yields

$$(2g_2+1)\alpha_3 + |\alpha_1-\alpha_2| < 1 \implies g_2\alpha_3 < \alpha_1. \tag{17}$$

Now, by (14), (15), (17) and the assumption $\alpha_1 \leqslant \alpha_2$, we get

$$\frac{5(1-\theta)}{6} \leqslant g_2\alpha_3 < \alpha_1 \leqslant \alpha_2 < 1-\theta. \tag{18}$$

Now Lemma 2.2 finishes the proof of the γ^2 part of Theorem 1, since

$$(2g_2-1)\alpha_3 < (4g_2-1)(2\theta-1) - \varepsilon^7 \leqslant (4g_2-1)(2\theta-1) + |\alpha_1-\alpha_2| - \varepsilon^7. \tag{19}$$

Next we shall prove Theorem 2. Assume that $\alpha_1 \leqslant \alpha_2$ as before. If the condition (i) holds, then we have

$$\alpha_1 = \frac{1-\alpha_3-|\alpha_1-\alpha_2|}{2} > \frac{1-\alpha_3-2\theta+1}{2} = 1-\theta-\frac{\alpha_3}{2} \geqslant \frac{5(1-\theta)}{8}, \tag{20}$$

and Case (i) of Theorem 2 follows easily by Lemma 2.3.

Now we can assume that $\frac{5(1-\theta)}{8} \leqslant \alpha_1 \leqslant \alpha_2 \leqslant 1-\theta$. Hence $\alpha_3 \geqslant 2\theta-1+|\alpha_1-\alpha_2| \geqslant 2\theta-1$. By Lemma 2.3, we only need to show that for some integer $k \geqslant 2$,

$$\mathcal{R}_{k\alpha_3}(w) > \frac{k(\alpha_3-(2\theta-1))}{2} + \varepsilon^7, \tag{21}$$

where

$$w = \begin{cases} w_1 = 1-\theta-\frac{1-\alpha_3-|\alpha_1-\alpha_2|}{5}, & \frac{5(1-\theta)}{8} \leqslant \alpha_1 \leqslant \frac{5(1-\theta)}{6}, \\ w_2 = \frac{3(1-\theta)-1+\alpha_3+|\alpha_1-\alpha_2|}{2}, & \frac{5(1-\theta)}{6} \leqslant \alpha_1 \leqslant 1-\theta. \end{cases}$$

Note that w_1 corresponds to Case (iii) and w_2 corresponds to Case (ii). Since $k(\alpha_3-(2\theta-1)) \geqslant 0$ and by (8), the condition (21) becomes

$$\frac{w}{2} > \frac{k(\alpha_3-(2\theta-1))}{2} + \varepsilon^7, \quad \text{and one of the following 3 conditions holds:} \tag{22}$$

$$\frac{w-(1-\theta)+2k\alpha_3}{6} > \frac{k(\alpha_3-(2\theta-1))}{2} + \varepsilon^7, \tag{23}$$

$$\frac{w-(1-\theta)+k\alpha_3}{2} > \frac{k(\alpha_3-(2\theta-1))}{2} + \varepsilon^7, \tag{24}$$

$$\frac{w-(1-\theta)+\frac{8}{5}k\alpha_3}{4} \quad \text{and} \quad \frac{w-\frac{1-\theta}{2}+k\alpha_3}{4} > \frac{k(\alpha_3-(2\theta-1))}{2} + \varepsilon^7. \tag{25}$$

Solving the inequalities (22)–(25) with $w = w_1$ and $w = w_2$, we get Case (ii) and Case (iii) of Theorem 2 with k replaced by g. Note that when $k\alpha_3 \geqslant 1-\theta$, we have $\frac{1}{k}\mathcal{R}_{k\alpha_3}(t) = \frac{t}{2k}$ decreases as k increases, hence $k > \max\left(2, \lceil \frac{1-\theta}{2\theta-1} \rceil\right)$ will not give any better results.

Finally, we show the γ^3 part of Theorem 1. Assume that $\alpha_1 \leqslant \alpha_2$ as before. Suppose that $\alpha_3 \leqslant \frac{3(1-\theta)}{4} - \varepsilon^5$, and thus (20) holds. If $\alpha_2 \geqslant 1-\theta$, then Case (i) of Theorem 2 gives the desired result. Hence, we assume that $\alpha_2 < 1-\theta$. The condition

$$\frac{5(1-\theta)}{8} \leqslant \alpha_1 \leqslant \alpha_2 \leqslant 1-\theta$$

in Lemma 2.3 is now satisfied, and we only need to show that (21) holds for some integer $k \geqslant 2$. Assume further that $\alpha_1 \geqslant \frac{5(1-\theta)}{6}$, which is equivalent to $\alpha_3 + |\alpha_1-\alpha_2| \leqslant \frac{5\theta-2}{3}$. Write $d = |\alpha_1-\alpha_2|$.

When $d = 0$, this reduces to $\alpha_3 \leqslant \frac{5\theta-2}{3}$. Now we can prove (21) under the assumption that (22) and (25) holds with $w = w_2$, and the 3 inequalities in (22) and (25) reduce to the 3 terms in $\gamma^3_{g_3}(\theta)$ with $k = g_3$. Write $t_1 = \frac{1-\theta}{2\theta-1}$ and take

$$k = \max\left(2,\ \left\lfloor \frac{5(t_1-1)}{12} \right\rfloor + 1\right). \tag{26}$$

Since $t_1 > 3$ for any $\frac{1}{2} < \theta < \frac{4}{7}$, we have

$$\frac{5t_1-5}{12} < k < \frac{t_1+1}{2}. \tag{27}$$

Putting (26) into (22) and (25), one can easily deduce that $\gamma^3_{g_3}(\theta) \geqslant 4\theta - 2 - \varepsilon$ for any $\frac{1}{2} < \theta < \frac{4}{7}$ by the condition (27) for k. Moreover, it is easy to verify that

$$\frac{3(1-\theta)}{4} \geqslant \gamma^3_{g_3}(\theta) \quad \text{and} \quad \frac{5\theta-2}{3} \geqslant \gamma^3_{g_3}(\theta) \tag{28}$$

for any $\frac{1}{2} < \theta < \frac{4}{7}$.

When $d > 0$, we can write $\alpha_1 = \frac{1-\alpha_3-d}{2}$ and $\alpha_2 = \frac{1-\alpha_3+d}{2}$. Note that the function $w(\alpha)$ has a slope $-\frac{2}{5}$ or -1, we have

$$w\left(\frac{1-\alpha_3-d}{2}\right) \geqslant w\left(\frac{1-\alpha_3}{2}\right). \tag{29}$$

Since $\mathcal{R}_{k\alpha_3}(t)$ is non-decreasing in $[\frac{\varepsilon^6}{2},\ 1-\theta]$, by (29) we have

$$\mathcal{R}_{k\alpha_3}\left(w\left(\frac{1-\alpha_3-d}{2}\right)\right) \geqslant \mathcal{R}_{k\alpha_3}\left(w\left(\frac{1-\alpha_3}{2}\right)\right) \tag{30}$$

for all $k \geqslant 2$. Thus, if $\mathcal{R}_{k\alpha_3}(w)$ satisfies (21) for $d = 0$, then it also satisfies (21) for all $0 < d < 2\theta - 1$. By the discussions in the $d = 0$ case, the proof of the γ^3 part of Theorem 1 is now completed.

AI declaration

ChatGPT-6 Astra was used in some numerical calculations and verifications in the proof of Theorem 1 and Theorem 2.

References

[1] R. C. Baker, G. Harman, and J. Pintz. The exceptional set for Goldbach's problem in short intervals. In *Sieve methods, exponential sums and their applications in number theory*, ed. G. R. H. Greaves, G. Harman and M. N. Huxley, Cambridge University Press, pages 1–54. Cambridge University Press, Cambridge, 1997.

[2] R. C. Baker, G. Harman, and J. Pintz. The difference between consecutive primes, II. *Proc. London Math. Soc.*, 83(3):532–562, 2001.

[3] B. Chen, V. Gupta, and Y. C. Li. Large value estimates for Dirichlet polynomials with characters and zero density of Dirichlet L-functions. *arXiv e-prints*, page arXiv:2507.08296v2, 2025.

[4] L. Guth and J. Maynard. New large value estimates for Dirichlet polynomials. *Ann. of Math.*, 203(2):623–675, 2026.

[5] G. Harman. *Prime-detecting Sieves*, volume 33 of *London Mathematical Society Monographs (New Series)*. Princeton University Press, Princeton, NJ, 2007.

[6] H. Iwaniec and E. Kowalski. *Analytic number theory*, volume 53 of *American Mathematical Society Colloquium Publications*. American Mathematical Society, Providence, RI, 2004.

[7] R. Li. The number of primes in short intervals and numerical calculations for Harman's sieve. *arXiv e-prints*, page arXiv:2308.04458v8, 2025.
[8] K. Matomäki, M. Pandey, J. Pliego, J. Teräväinen, and M. Wang. Higher order uniformity of the primes and cancellation of the Möbius function in shorter intervals. *arXiv e-prints*, page arXiv:2610.09567v1, 2026.

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